ARYAN HIRLEKAR

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EDUCATION

Master of Science in Computer Science **Sep 2025 - May 2027 (Expected)**
Northeastern University, Boston GPA: 3.6/4
Coursework: Programming Design Paradigms, Algorithms, Database Management Systems, Foundations of AI

Bachelor of Technology in Computer Science **Jul 2021 - Jun 2025**
Fr. Conceicao Rodrigues Institute of Technology GPA: 3.4/4

EXPERIENCE

AI/ML Intern **Jul 2026 - Dec 2026**
Resolve Tech Solutions / Juno Labs AI Addison, TX

- Built the frontend generation agent for a multi-agent SDLC platform on AWS Bedrock AgentCore; engineered cross-layer consistency guardrails ensuring generated React, FastAPI, and Postgres layers share contracts, eliminating API drift between agents and validated across **50+ generated applications**
- Deployed the agent as a standalone AgentCore runtime (Docker, ECR, A2A JSON-RPC) with S3 artifact handoff; cut downstream prompt payloads by replacing the **17KB+** OpenAPI spec embedded in every agent prompt with pointer references and prompt caching
- Built the monitoring, analysis, root cause, and ticket agents in a **9-agent** LangGraph orchestration graph on Juno HawkAI V2, detecting AWS anomalies, running RCA via Bedrock Claude Sonnet 4, and raising enriched incidents in ServiceNow and Jira via a pluggable MCP ticketing server
- Implemented ticket enrichment for unseen incident types by routing unknown anomalies through an AgentCore web search gateway, automatically surfacing remediation steps and appending them as structured work notes in the ServiceNow ticket

AI/ML Intern **Jun 2024 - Sep 2024**
Tech Mahindra Maker's Lab Pune, India

- Built and deployed a sentence-embedding microservice in PyTorch achieving **85%** semantic similarity accuracy
- Designed and fine-tuned a multi-class classifier on a 15,000-sample corpus, lifting accuracy from **71% to 88%** across 6 categories via hyperparameter tuning

PROJECTS

RiskFlow - Event-Driven Transaction Risk Scoring Platform **Spring Boot, Kafka, Redis, Postgres, Docker**

- Architected a 4-stage asynchronous microservice pipeline (rules engine, velocity analytics, hot-reloadable rules, enrichment) delivering sub-second verdicts with full audit logging via Spring Boot and Kafka
- Sustained **200 TPS (12k transactions/min)** over a 5-minute load test at **p95 37ms** with **99.975%** end-to-end pipeline success across 60,000 transactions; cut rule deployment downtime to **zero** via hot reload
- Instrumented the platform with OpenTelemetry distributed tracing (**151-span** cross-service traces through Kafka, verified in Jaeger), Testcontainers integration tests, and a Redis-backed distributed rate limiter

BlueprintAI - Agentic Engineering Drawing Pipeline **React, Node.js, Python, REST APIs, Docker**

- Designed a dual-parser backend (pdfplumber for text PDFs, PyMuPDF for scans) achieving **92%** field extraction accuracy across 200+ line items per job
- Exposed extracted part data to LLM agents and downstream systems through MCP tools (get_part_specs, get_dimensions, reconcile_bom), with per-field confidence scoring and a human-in-the-loop review UI for low-confidence fields

PitStrat AI - F1 Race Strategy Simulator **FastAPI, React, XGBoost, IBM Granite, MCP**

- Trained an XGBoost tyre-degradation model on **71,999 lap records** across 72 races (**MAE 1.95s**, **87%** variance explained); powered a counterfactual engine validated on the 2025 Qatar GP (P2 to P1 with a safety-car pit)
- Exposed the engine through 5 typed MCP tools called by IBM Granite (watsonx.ai), enabling natural-language strategy queries; won **1st place** at the IBM SkillsBuild AI Builders Challenge (May 2026)

SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C++
Frameworks: Spring Boot, React, Node.js, Express, Flask, FastAPI, REST APIs
Testing & Observability: Testcontainers, JUnit, pytest, OpenTelemetry, Jaeger, Resilience4j, AWS CloudWatch
Systems & Infra: Apache Kafka, Redis, Docker, Docker Compose, Microservices, Terraform, AWS (ECR, S3, IAM), Git, GitHub Actions, CI/CD
Databases & Integrations: PostgreSQL, MySQL, MongoDB, ServiceNow, Jira

AWARDS, PUBLICATIONS & CERTIFICATIONS

- **1st Place, IBM SkillsBuild AI Builders Challenge** **May 2026**
- **Personalised Module Recommendation System Using AI** - ICCCNT 2025, IEEE Xplore (in press) **Jul 2025**
- **Model Context Protocol: Advanced Topics** | Anthropic **Mar 2026**
- **Introduction to Model Context Protocol** | Anthropic **Mar 2026**